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Convergence Tests for Electronics and Telecommunications

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Number series $\sum_{n=1}^{\infty} a_n$

Given a sequence of numbers (a_k), $a_k \in \mathbb{R}$, indexed by natural numbers k we define a **number series** as a sequence of **partial sums** (s_n), $s_n := \sum_{k=1}^{n} a_k$.
If the limit of partial sums exists and is finite

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty}(a_0 + a_1 + \ldots + a_n)$$

the series $\sum_{n=1}^{\infty} a_n$ is called **convergent**, otherwise ie. if it does not exist or is equal $\pm\infty$ it is called **divergent**, or divergent to $\pm\infty$, respectively.
Instead of writing $\sum_{n=1}^{\infty} a_n$, for short we shall write $\sum a_n$.

Absolute convergence versus convergence

If the series $\sum_{n=1}^{\infty} a_n$ is convergent and the series of absolute values $\sum_{n=1}^{\infty} |a_n|$ is convergent, then the series is called **absolutely convergent**.

If $\sum_{n=1}^{\infty} a_n$ is convergent, but $\sum_{n=1}^{\infty} |a_n|$ is divergent, then the series is called **conditionally convergent**.

The series of the form $\sum_{n=1}^{\infty} (-1)^n a_n$ is called an **alternating series**.

Cauchy's condition

Definition of the Cauchy's condition

A number series $\sum a_n$ satisfies Cauchy's condition if

$$\forall \varepsilon > 0 \ \ \exists\, N \in \mathbb{N} \ \ \forall\, r, t \geqslant N \ \ : |a_r - a_t| < \varepsilon.$$

Necessary and sufficient condition of convergence

The series $\sum a_n$ is convergent iff it satisfies Cauchy's condition.

Necessary condition of convergence

If the series $\sum a_n$ is convergent, then $a_n \to 0$, as n is approaching ∞.

Notice that from the necessary condition we learn that $\sum (-1)^n$ is divergent.

The Majorant/Weierstrass' Test

Assume $\sum b_n$ is the upper bound for a series $\sum a_n$

$$\text{i.e. } |a_n| < b_n, \text{ for all } n \geqslant n_0 \text{ for same } n_0$$

and assume that $\sum b_n$ is convergent.
Then $\sum a_n$ and $\sum |a_n|$ are convergent, ie. the series $\sum a_n$ is absolutely convergent.

D'Alambert's Ratio Test

If for a series $\sum a_n$

a) $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1 \Rightarrow \sum a_n -$ convergent

b) $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1 \Rightarrow \sum a_n -$ divergent

If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$ the test is inconclusive.

Instead of the limit one can check if expression $\left|\frac{a_{n+1}}{a_n}\right|$ is smaller, or greater then some number q, respectively $q < 1$ for (a), $q > 1$ for (b).

D'Alambert's Ratio Test

Example
Check convergence of $\sum \frac{n!}{n^n}$.

$$\frac{a_{n+1}}{a_n} = \frac{n!(n+1)}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)(n+1)^n} = \frac{1}{\frac{(n+1)^n}{n^n}} = \frac{1}{\left(1+\frac{1}{n}\right)^n} \to \frac{1}{e}$$

.
Since $\frac{1}{e} < 1$,
from Ratio Test series $\sum \frac{n!}{n^n}$ is convergent.

Cauchy's Root Test

If for a series $\sum a_n$

a) $\lim_{n\to\infty} \sqrt[n]{|a_n|} < 1 \Rightarrow \sum a_n$ convergent

b) $\lim_{n\to\infty} \sqrt[n]{|a_n|} > 1 \Rightarrow \sum a_n$ divergent

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$ and the and the limit approaches strictly from above then the series diverges, otherwise the test is inconclusive.
Instead of the limit, one may use limit superior in the test.

Cauchy's Root Test

Example

Check convergence of $\sum_{n=0}^{\infty} a_n$, where $a_{2n} = \frac{1}{2^{2n}}$, $a_{2n+1} = \frac{3}{2^{2n+1}}$

$$\sqrt[2n]{\frac{1}{2^{2n}}} = \left(\frac{1}{2^{2n}}\right)^{\frac{1}{2n}} = \frac{1}{2} < 1$$

$$\sqrt[2n+1]{\frac{3}{2^{2n+1}}} = \left(\frac{3}{2^{2n+1}}\right)^{\frac{1}{2n+1}} = \frac{3^{\frac{1}{2n+1}}}{2} \to \frac{1}{2} < 1$$

From the Root Test we obtain that $\sum_{n=0}^{\infty} a_n$ is convergent.

Comparison of d'Alambet's Ratio Test with Cauchy's Root Test

The Root Test is stronger then the Ratio Test, so it means that the Ratio Test can be inconclusive, but the Root Test will provide the answer.

Root Test is stronger then Ratio Test

Example showing that the Root Test is stronger then The Ratio Test

Consider the following series:

$$\sum a_n = \underset{a_0}{1} + \underset{a_1}{\frac{3}{2}} + \underset{a_2}{\frac{1}{2^2}} + \underset{a_3}{\frac{3}{2^3}} + \underset{a_4}{\frac{1}{2^4}} + \dots + \frac{1}{2^{2n}} + \frac{3}{2^{2n+1}} + \dots$$

where $a_{2k} = \frac{1}{2^{2k}}$, $a_{2k+1} = \frac{3}{2^{2k+1}}$

Let's try the Ratio Test

$$\frac{a_{2k+1}}{a_{2k}} = \frac{3 \cdot 2^{2k}}{2^{2k} \cdot 2} = \frac{3}{2} > 1$$

$$\frac{a_{2k}}{a_{2k-1}} = \frac{2^{2k-1}}{2^{2k} \cdot 3} = \frac{2^{2k}}{2^{2k} \cdot 2 \cdot 3} = \frac{1}{6} < 1$$

inconclusive (it doesn't give the answer)

The Root Test

$$\sqrt[2k]{\frac{1}{2^{2k}}} = \frac{1}{2} < 1 \; ; \; \sqrt[2k+1]{\frac{3}{2^{2k+1}}} = \frac{3^{\frac{1}{2k+1}}}{2} \to \frac{1}{2} < 1 \to \text{convergent}$$

Integral Test

Integral Test

Let $f : [1, +\infty) \to \mathbb{R}$ be monotonically decreasing function, such that $f(n) = a_n$. The series $\sum s_n$ is convergent iff

$$\int_1^{+\infty} f(x)\mathrm{dx} \text{ is finite.}$$

Harmonic series $\sum \frac{1}{n^\alpha}$, with constant $\alpha \in (0; \infty)$

Convergence of a harmonic series depends on α

If $0 < \alpha \leqslant 1$, then $\sum \frac{1}{n^\alpha}$ is divergent.
If $\alpha > 1$, then $\sum \frac{1}{n^\alpha}$ is convergent.

Examples
$\sum \frac{1}{n^2}, \sum \frac{1}{n^{\frac{3}{2}}}$ is convergent.
$\sum \frac{1}{\sqrt{n}}, \sum \frac{1}{n}$ are divergent.

Example of a Convergent Harmonic Series

Example.
Let's check if $\sum \frac{1}{k^2}$ is convergent, by compering it with $\sum \frac{1}{(k-1)k}$.

$$\frac{1}{k^2} < \frac{1}{(k-1)k} = \frac{1}{k-1} - \frac{1}{k}$$

$$s_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \ldots + \left(\frac{1}{n-1} - \frac{1}{n}\right) =$$

$$= 1 - \frac{1}{n}.$$

$\lim_{n\to\infty}(1 - \frac{1}{n}) = 1.$

Implementing Integral Test for a Harmonic Series

• Example of a convergent series.
The series $\sum \frac{1}{n^2}$ is convergent, because $f(x) = \frac{1}{x^2}$ is a monotonically decreasing function and

$$\int_1^{+\infty} \frac{1}{x^2} \mathrm{dx} = \lim_{\mathrm{b} \to +\infty} -\frac{1}{\mathrm{x}}\Big|_1^{\mathrm{b}} = \lim_{\mathrm{b} \to +\infty} -\frac{1}{\mathrm{b}} + 1 = 1 < +\infty$$

is finite.
•Example of a divergent series.
The series $\sum \frac{1}{\sqrt{n}}$ is divergent, because $f(x) = \frac{1}{\sqrt{x}}$ is a monotonically decreasing function and

$$\int_1^{+\infty} \frac{1}{\sqrt{x}} \mathrm{dx} = \lim_{\mathrm{b} \to +\infty} 2\sqrt{\mathrm{x}}\Big|_1^{\mathrm{b}} = \lim_{\mathrm{b} \to +\infty} 2\sqrt{\mathrm{b}} - 2 = +\infty.$$

Example with harmonic series as a majorant

Check convergence of $\sum_{n=1}^{\infty} \frac{\left(1+\frac{1}{n}\right)^n}{(n+1)\sqrt{2n+1}}$.

$$\sum_{n=1}^{\infty} \frac{\left(1+\frac{1}{n}\right)^n}{(n+1)\sqrt{2n+1}} < \sum \frac{3}{n\sqrt{n}} = 3\sum n^{-\frac{3}{2}}$$

And the original series is convergent, since its majorant is a harmonic series with $\alpha = \frac{3}{2} < 1$ so it is convergent.

Leibniz Test

Convergence Test for an Alternating Series

If a_n decreases to monotonically to 0, then $\sum(-1)^n a_n$ is convergent.

Example.

$$\sum \frac{(-1)^n}{\sqrt{n}} \quad \text{convergent}$$

Check convergence of the following series

1) $\sum \left(1-\frac{1}{n}\right)^n$
2) $\sum \frac{1}{3n+1}$
3) $\sum \frac{1}{n^2+n}$
4) $\sum \frac{3^n-2^n}{4^n-3^n}$
5) $\sum \frac{2n}{n!}$
6) $\sum \frac{(n!)^3}{(2n)!}$
7) $\sum \left(\frac{n+2}{n+3}\right)^n$
8) $\sum \frac{(-2)^n}{n^2}$
9) $\sum \sqrt[n]{\frac{n}{1000}}$
10) $\sum \frac{1}{n\cdot \ln n}$
11) $\sum \frac{1}{\sqrt{n}+1}$
12) $\sum \left(1-\cos\left(\frac{1}{n}\right)\right)$
13) $\sum \frac{2^n+3^n}{3^n+4^n}$
14) $\sum (\arccos\left(\frac{1}{n}\right))^n$
15) $\sum (-1)^n \cdot \frac{1}{n!}$
16) $\sum \frac{n}{\ln(n)}$
17) $\sum n \cdot \sin\left(\frac{1}{n^3}\right)$
18) $\sum \frac{1}{\sqrt{n(n+1)}}$
19) $\sum \ln^n(2+\frac{1}{n})$
20) $\sum (-1)^n \cdot \frac{n}{4^n}$
21) $\sum \frac{\ln n}{\pi^n}$